

Trigonometry

$(\cos \theta, \sin \theta)$ is the coordinate on the unit circle that makes angle θ with the positive x -axis.

$$\sec \theta = \frac{1}{\cos \theta} \quad \csc \theta = \frac{1}{\sin \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

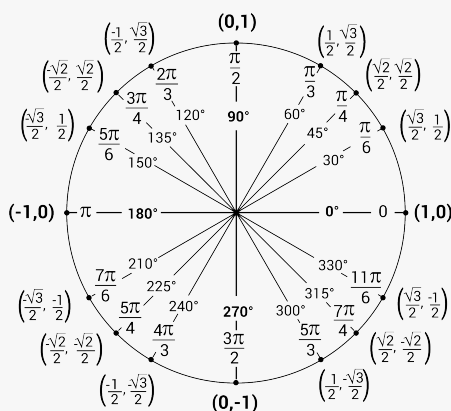
Pythagorean identities $\begin{cases} \sin^2 \theta + \cos^2 \theta = 1 \\ \tan^2 \theta + 1 = \sec^2 \theta \\ 1 + \cot^2 \theta = \csc^2 \theta \end{cases}$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$



Limits

Law Let $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$.

Sum $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$

Scalar $\lim_{x \rightarrow a} cf(x) = cL$

Product $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = L \cdot M$

Quotient $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$ for $M \neq 0$

Power $\lim_{x \rightarrow a} (f(x))^n = L^n$

Root $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}$ for all L if n is odd, and for $L \geq 0$ if n is even and $f(x) \geq 0$.

Squeeze Theorem:

Let f, g , and h be functions with $g(x) \leq f(x) \leq h(x)$ for all x and $\lim_{x \rightarrow a} g(x) = L = \lim_{x \rightarrow a} h(x)$, then $\lim_{x \rightarrow a} f(x) = L$.

Indeterminate Forms:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0^0, \infty - \infty, 1^\infty, 0 \cdot \infty, \infty^0$$

$\epsilon - \delta$ definition:

L is the limit of f as x approaches a if for all $\epsilon > 0$, there is some $\delta > 0$, such that $0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$.

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1 \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \lim_{x \rightarrow \infty} \frac{ax^n + \dots}{bx^m + \dots} = \begin{cases} 0 & m > n \\ \infty & n > m \\ a/b & n = m \end{cases}$$

Continuity

Definition: f is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$. If f is defined near a (except possibly at a itself) but fails the continuity condition, then f is discontinuous at a .

- polynomials, trig (and inverses), exponentials, and logs are **CTS on their domains**
- The algebraic combination of continuous functions is continuous.

Composite Function Theorem:

If $f(x)$ is continuous at L and $\lim_{x \rightarrow a} g(x) = L$, then $\lim_{x \rightarrow a} f(g(x)) = f(L)$.

Intermediate Value Theorem:

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$.

Finding Derivatives

Limit definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Tangent line to $f(x)$ at $x = a$:

$$L(x) = f(a) + f'(a)(x - a)$$

L'Hôpital's Rule:

If f, g are differentiable on a deleted neighborhood of a , $g' \neq 0$ there, f/g has form $0/0$ or ∞/∞ , and $L = \lim_{x \rightarrow a} f'(x)/g'(x)$ exists (possibly infinite), then $\lim_{x \rightarrow a} f(x)/g(x) = L$.

Logarithmic Differentiation ($f(x) > 0$):

To find the derivative of $y = f(x)^{g(x)}$, take $\ln()$ of both sides, bring $g(x)$ down using the log rule ($\ln(a^b) = b \ln(a)$):

$$\ln(y) = \ln(f(x)^{g(x)}) = g(x) \ln(f(x))$$

Then implicitly differentiate and solve for y' :

$$y' = f(x)^{g(x)} \left(g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)} \right).$$

Power Rule

$$[x^a]' = ax^{a-1}$$

Trig Rules

$$[\sin(x)]' = \cos(x)$$

$$[\cos(x)]' = -\sin(x)$$

(PSST!)

$$[\tan(x)]' = \sec^2(x)$$

$$[\cot(x)]' = -\csc^2(x)$$

$$[\sec(x)]' = \sec(x) \tan(x)$$

$$[\csc(x)]' = -\csc(x) \cot(x)$$

Inverse Trig Rules

$$[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$$

$$[\arccos(x)]' = \frac{-1}{\sqrt{1-x^2}}$$

$$[\arctan(x)]' = \frac{1}{1+x^2}$$

$$[\text{arccot}(x)]' = \frac{-1}{1+x^2}$$

$$[\text{arcsec}(x)]' = \frac{1}{|x|\sqrt{x^2-1}}$$

$$[\text{arccsc}(x)]' = \frac{-1}{|x|\sqrt{x^2-1}}$$

Exponent Rule

$$[a^x]' = \ln(a)a^x$$

Logarithm Rule

$$[\log_a(x)]' = \frac{1}{x \ln(a)}$$

Scalar Rule $[af]' = af'$

Sum Rule $[f+g]' = f' + g'$

Product Rule $[fg]' = f'g + fg'$

Quotient Rule $\left[\frac{f}{g}\right]' = \frac{f'g - fg'}{g^2}$

Chain Rule $[f(g(x))]' = f'(g(x))g'(x)$

Inverse Rule $[f^{-1}(x)]' = \frac{1}{f'(f^{-1}(x))}$

Requires f differentiable and one-to-one, $f'(f^{-1}(x)) \neq 0$.

Definitions

- The *definite integral* of f on (a, b) is written $\int_a^b f(x) dx$ and is defined to be the *signed* area between the graph of f and the x -axis (if such a quantity exists).
- The *indefinite integral* (or *anti-derivative*) of f on is written $\int f(x) dx$ or $\int f$ is the family of functions whose derivative is f .

Fundamental Theorem of Calculus If f is continuous on $[a, b]$ and F is continuous on $[a, b]$ with $F' = f$ on (a, b) , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Scalar Rule

$$\int af = a \int f.$$

Sum Rule

$$\int (f + g) = \int f + \int g$$

Int. by Parts

$$\int f'g = fg - \int fg'$$

u-substitution

$$\int f'(g(x))g'(x) dx = f(g(x))$$

Power Rule

$$\int x^a dx = \begin{cases} \frac{1}{a+1}x^{a+1} + C & a \neq -1 \\ \ln|x| + C & a = -1 \end{cases}$$

Trig Rules

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$

Exp. Rules

$$\int a^x dx = \frac{a^x}{\ln(a)} + C, \quad a > 0, a \neq 1$$

Partial Fractions

Factor	Term in decomposition
$ax + b$	$\frac{A}{ax+b}$
$(ax + b)^k$	$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_k}{(ax+b)^k}$
$ax^2 + bx + c$	$\frac{Ax+B}{ax^2+bx+c}$
$(ax^2 + bx + c)^k$	$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_kx+B_k}{(ax^2+bx+c)^k}$

Trig Substitutions

Integrand	Substitution	Result
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$a \sec \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$a \tan \theta$

Assume $a > 0$. Use $|\theta| \leq \pi/2$ for sine and $|\theta| < \pi/2$ for tangent. For secant, use $0 \leq \theta < \pi/2$ if $x \geq a$, or $\pi \leq \theta < 3\pi/2$ if $x \leq -a$.

Other Antiderivatives

$\int \ln(x) dx = x \ln(x) - x + C$
$\int \tan(x) dx = -\ln \cos(x) + C$
$\int \sec(x) dx = \ln \sec(x) + \tan(x) + C$
$\int \csc(x) dx = \ln \csc(x) - \cot(x) + C$
$\int \cot(x) dx = \ln \sin(x) + C$

Riemann Sums

$$R_n = \sum_{k=1}^n f\left(a + k \frac{b-a}{n}\right) \frac{b-a}{n}$$

$$L_n = \sum_{k=1}^n f\left(a + (k-1) \frac{b-a}{n}\right) \frac{b-a}{n}$$

$$T_n = \sum_{k=1}^n \frac{f\left(a + (k-1) \frac{b-a}{n}\right) + f\left(a + k \frac{b-a}{n}\right)}{2} \frac{b-a}{n}$$

$$\lim_{n \rightarrow \infty} R_n, L_n, T_n = \int_a^b f(x) dx$$

Sums of Powers

- $\sum_{k=1}^n 1 = n$
- $\sum_{k=1}^n k = \frac{n(n+1)}{2}$
- $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$

Power Series

Near $x = a$,

$$f(x) \approx T_k(x) = \sum_{n=0}^k \frac{f^{(n)}(a)}{n!} (x-a)^n,$$

and on some interval centered at a ,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Function	Taylor Series	Interval of Convergence
e^x	$\sum_{n=0}^{\infty} \frac{1}{n!} x^n$	$\mathbb{R}$
$\sin(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$	$\mathbb{R}$
$\cos(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$	$\mathbb{R}$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$(-1, 1)$
$\ln(1-x)$	$\sum_{n=1}^{\infty} \frac{-1}{n} x^n$	$[-1, 1)$
$(x+1)^k$	$\sum_{n=0}^k \binom{k}{n} x^n$	$\mathbb{R} \ (k \in \mathbb{Z}_{\geq 0})$

Test for Convergence and Divergence

- Divergence Test:** $\lim_{n \rightarrow \infty} a_n \neq 0 \implies \sum a_n$ diverges.
- Integral Test:** If $f(x) > 0$ is CTS and decreasing on $[k, \infty)$, and $f(n) = a_n$,

$$\int_k^{\infty} f(x) dx \text{ converges} \iff \sum_{n=k}^{\infty} a_n \text{ converges.}$$

- The p -series Test:** If $k > 0$, $\sum_{n=k}^{\infty} \frac{1}{n^p}$ converges $\iff p > 1$.
- Comparison Test:** If $0 \leq a_n \leq b_n$ for all n , then

$$\sum b_n \text{ converges} \implies \sum a_n \text{ converges.}$$

- Limit Comparison Test:** If $a_n \geq 0$ and $b_n > 0$ for all n and $\lim_{n \rightarrow \infty} a_n/b_n$ is positive and finite, then

$$\sum a_n \text{ converges} \iff \sum b_n \text{ converges.}$$

- Alternating Series Test:** If $b_n \geq 0$ decreases to 0, $\sum b_n(-1)^n$ converges.
- Absolute Convergence Test:**

$$\sum |a_n| \text{ converges} \implies \sum a_n \text{ converges.}$$

- Ratio Test:** Define $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$. If $L < 1$, $\sum a_n$ converges. If $L > 1$, $\sum a_n$ diverges.
- Root Test:** Define $L = \lim_{n \rightarrow \infty} |a_n|^{1/n}$. If $L < 1$, $\sum a_n$ converges. If $L > 1$, $\sum a_n$ diverges.

Bounding Error

If $f(x) > 0$ is CTS and decreasing on $[1, \infty)$ with $f(n) = a_n$, then for any N ,

$$\left(\sum_{n=1}^N a_n + \int_{N+1}^{\infty} f(x) \, dx\right) \leq \sum_{n=1}^{\infty} a_n \leq \left(\sum_{n=1}^N a_n + \int_N^{\infty} f(x) \, dx\right).$$

Suppose $S = \sum_{n=1}^{\infty} b_n(-1)^n$ where $b_n \geq 0$, and b_n decreases to 0. Then for any N ,

$$|S - S_N| \leq b_{N+1}$$

where $S_N = \sum_{n=1}^N b_n(-1)^n$.

Taylor’s Inequality: If for all $x \in [a - d, a + d]$,

$$\left|f^{(k+1)}(x)\right| \leq M,$$

then for all $x \in [a - d, a + d]$,

$$|f(x) - T_k(x)| \leq \frac{M}{(k + 1)!} |x - a|^{k+1}.$$

Solving ODEs

To solve the separable ODE $\frac{dy}{dx} = f(x)g(y)$, separate the x and y variables and integrate:

$$\int \frac{1}{g(y)} \, dy = \int f(x) \, dx.$$

To solve a 2nd order ODE of the form $ay'' + by' + cy = 0$ with $a \neq 0$, find the roots λ of the characteristic equation

$$a\lambda^2 + b\lambda + c = 0.$$

Case	Roots	Solutions
$\Delta > 0$	λ_1, λ_2	$y = pe^{\lambda_1 x} + qe^{\lambda_2 x}$
$\Delta = 0$	$\lambda = -b/(2a)$	$y = (p + qx)e^{\lambda x}$
$\Delta < 0$	$\alpha \pm i\beta$	$y = e^{\alpha x}(p \cos(\beta x) + q \sin(\beta x))$

Here $\Delta = b^2 - 4ac$, $\alpha = -b/(2a)$, and $\beta = \sqrt{4ac - b^2}/(2|a|)$.

Common ODEs

Name	ODE	Solution
Exponential	$y' = ky$	$y = e^{kx+C}$
Logistic	$y' = ky \left(1 - \frac{y}{M}\right)$	$y = \frac{M}{1 \pm Ce^{-kx}}$
Harmonic Oscillator	$-ky = my''$	$y = A \cdot \cos\left(\sqrt{\frac{k}{m}}x + C\right)$